“Prediction of Device Choice from Behavioral Segments”

was demonstrated by using the dataset (/kaggle/input/lifestyle-dataset/genz_digital_lifestyle_tech_habits.csv). The dataset contained 500 rows representing real-world digital lifestyle of GenZ. The dataset was synthetic and created for research and analysis purposes. The variables used for this project were User_ID, Age, Gender, Daily_Screen_Time_hrs, Preferred_Device, Social_Media_Usage_hrs, Gaming_Activity, Tech_Savviness, Streaming_Subscriptions and Privacy_Awareness_Level. All analyses were carried out in Python.

There were no missing values, no strong outliers, and no marked skewness. The order was mapped from low to high for the ranked categories. To perform the prediction, the variables were separated into the Features (X) and the Target (y). Preferred_Device and User_ID were dropped from X and Preferred_Device was assigned to y.

The analysis utilized a **Scikit-Learn Pipeline** that automates the transition from raw data to predictions, ensuring that preprocessing steps are applied consistently and preventing data leakage. The following are the three primary stages of the pipeline:

1. Feature Transformation

A **Column Transformer** was employed to apply type-specific logic to the dataset’s variables simultaneously.

- **Ordinal Encoding**: For features with a natural hierarchy (Gaming_Activity, Tech_Savviness, Privacy_Awareness), an ordinal encoder was used with manual mapping. This preserves the meaningful distance between levels rather than treating them as arbitrary categories.
- **Nominal Encoding**: For non-ranked categories (Gender), One Hot Encoder was used. By setting *handle_unknown=”ignore”*, the pipeline was made more robust against future data discrepancies, while *sparse_output=False* ensured compatibility with the subsequent scaling step.
- Passthrough: Numerical columns were passed through without initial modification, maintaining their original integrity before feature scaling.

2. Feature Scaling

Following encoding, a MinMaxScaler was applied to the entire feature set. This normalized all variables to a consistent range (0 to 1), ensuring that numerical features did not disproportionately influence the model due to their original scale.

3. Predictive Modeling with Regularization

The final stage of the pipeline utilizes a Random Forest Classifier.

To check the reliability of the model, **cross-validation** was done with cv=5. Model reliability was as follows:

- Accuracy per fold: [0.21 0.21 0.22 0.16 0.23]
- Average accuracy: 0.21
- Stability (std Dev): 0.02

Given that the model accuracy was 21%, the model was configured with specific constraints that prioritize simplification:

- Max Depth (5): Limits the complexity of the decision trees to prevent them from over-fitting to noise.
- Minimum Samples per Leaf (15): Ensures that every logical "path" the model finds is backed by a statistically significant group of at least 15 individuals, reinforcing the reliability of the findings.

With the constraints, the accuracy moved from 0.21 to 0.22, and the stability (Standard Deviation) increased from 0.02 to 0.04. Accuracy per fold was [0.22 0.24 0.15 0.22 0.27]. This showed that the model was still guessing. With 5 device types, 20% was the 'random baseline'. Being at 22% means that even after simplification, the current features were not strong enough to predict device choice for the 500 people.

To see which variables actually drove the predictions, **'Feature Importance'** was performed. While the figure was not the 'Flat' or the 'One-Hit Wonder', the top features 'Daily Screen Time' and 'Social Media Usage' were just around 20%.

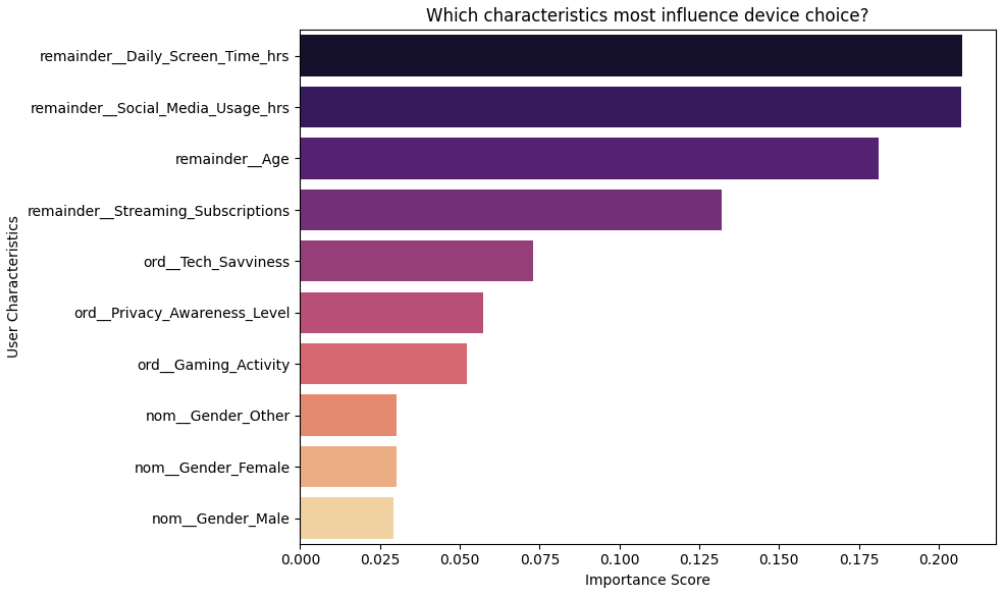


Figure (1) Importance Score Vs User Characteristics

To see where the model was confused, a **confusion matrix** was performed. It highlighted varying levels of predictive accuracy across the five device categories. The model was most accurate at identifying Tablets, with 82% correct predictions. It really struggled with Laptops – only 19 were correctly identified. As the model misidentified 33 Laptops as Tablets, there was a major overlap between Laptops and Tablets.

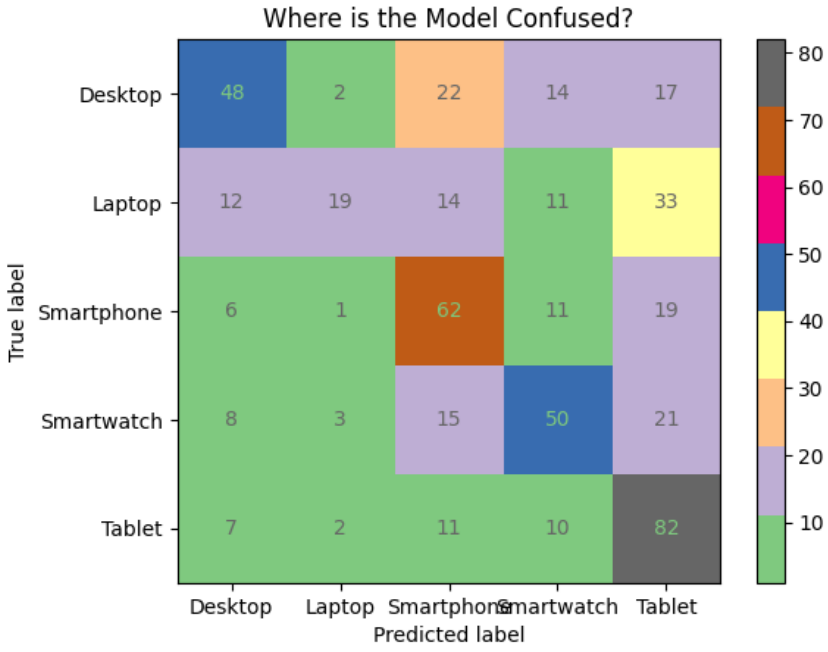


Figure (2) Confusion Matrix (Five device categories)

So, instead of 5 device types, they were grouped into three – Desktop as 'Fixed', Laptop and Tablet as 'Large-Portable', and Smartphone and Smartwatch as 'Small-Portable' to

mitigate the model's confusion. With class_weight='balanced', the pipeline and the cross validation were re-run. The average accuracy became 0.35.

Given the new grouping, the **'Feature Importance'** showed that 'Daily Screen Time' was the top feature with about 25% importance and 'Gender_Female' with about 2.5% importance. As the scores were significantly different from each other, it can be assumed that the model was trying to use that column as its primary anchor. However, with only 35% of average accuracy, it was still not enough to accurately sort people into the three groups on its own.

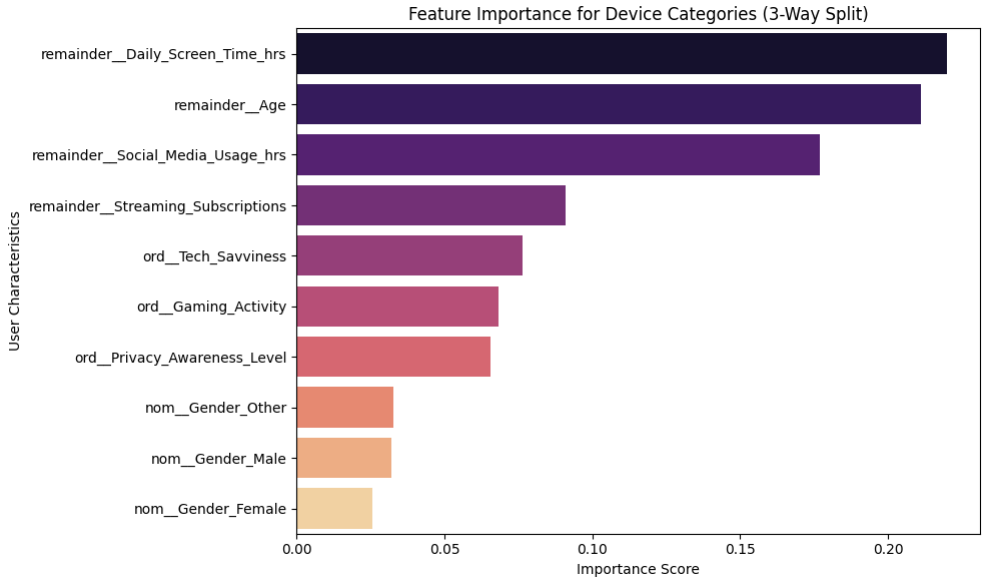


Figure (3) Importance Score (3 Device Categories) Vs User Characteristics

Since 'Daily Screen Time' was the top feature, it suggests that the groups were defined by how much they use technology, not by who they are (Gender) or what they know (Tech Savviness). Meanwhile, the reason the accuracy (35%) was not higher was likely that higher screen time was shared across too many devices. For example, a frequent gamer on a desktop and a frequent social media user on a smartphone both had high screen time, so the model cannot distinguish them based on that alone.

So, as a final attempt, I tried **Interaction Features**. 'Tech_Intensity' was created by multiplying Daily_Screen_Time_hrs and Tech_Savviness, and 'Gaming_Intensity' was created by multiplying Daily_Screen_Time_hrs and Gaming_Activity. After updating X and y, the pipeline was re-run, and the cross validation was checked. The average accuracy was 0.36, and the standard deviation was 0.05.

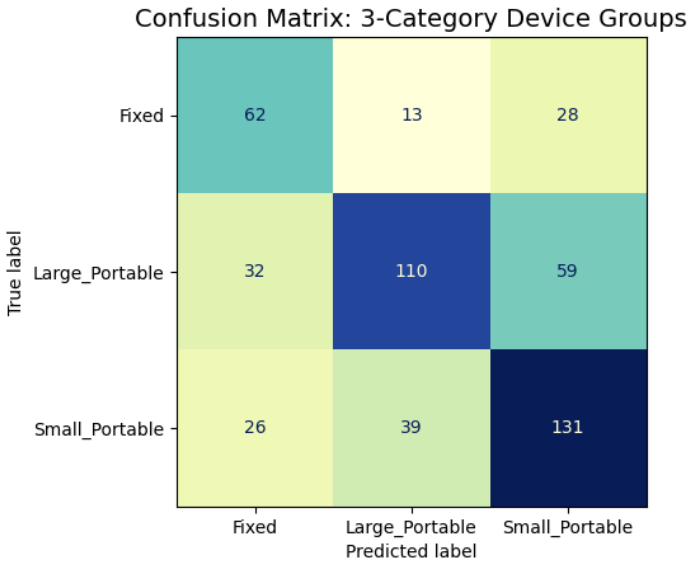


Figure (4) Confusion Matrix (3-Category Device Groups)

Next, the **confusion matrix** was done, and the result showed that there was a significant 'leakage' between the Large_Portable and the Small_Portable. This suggests that the updated features might not clearly distinguish between different sizes of portable devices. To visualize this 'Similarity Trap', a **distribution plot** was created. Figure (5) shows that all three curves occupied almost the exact same space on the horizontal axis (from roughly 2 to 14 hours). For example, if a person had 7.5 hours of screen time, they could realistically fall into any of the three categories.

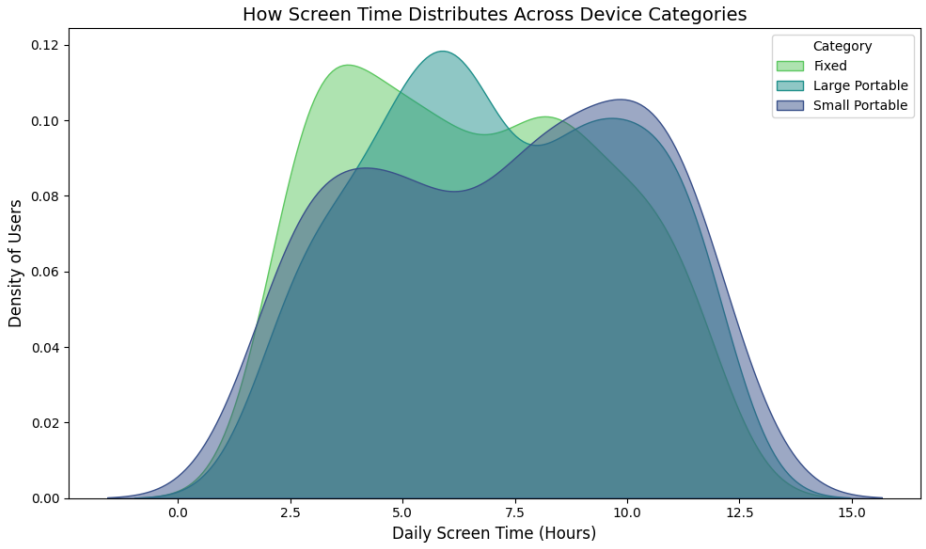


Figure (5) Distribution Plot showing Screen Time Distribution across Device Categories

All in all, with a jump from 0.35 to 0.36 and a higher Standard Deviation (0.05), the model reached the 'mathematical ceiling' for this dataset. A score of 0.36 (where 0.33 was random guessing) means that the features – even with clever math – were almost entirely unrelated to the device people chose. So, it can be assumed that while device

choice appeared highly individualized in this population, Daily_Screen_Time was the ‘least random’ predictor.

Appendix

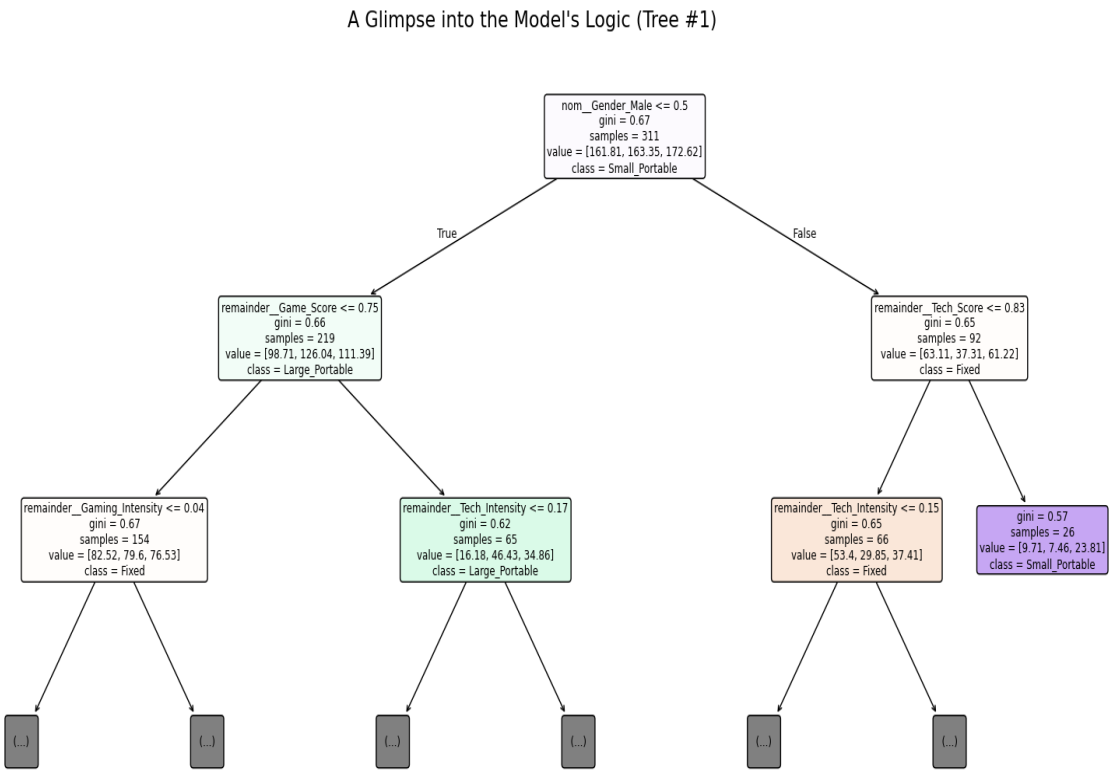


Figure (6) An example Tree of Random Forest (Tree no. 1 was presented as an example. Gender_Male was used as the first split condition and the sample size, the random subset of sample, was 311. Gini was around 0.6, indicating the significant mixed nature of the classes.)